On-blade induced drag of curved lifting lines in the Vortex Step Method: problem statement, literature, and a parameter-free regularisation

Working document, awegroup/Vortex-Step-Method, `develop`

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Abstract

The `develop` branch of the Python Vortex Step Method (VSM) was aligned with the coupling rules of Gaunaa, Li and Pirrung (TORQUE 2026): force magnitudes from the three-quarter-chord flow, force directions from the quarter-chord flow, and a Kutta–Joukowski force on the attached-trailed vortex legs. With these rules the on-blade induced drag equals the Trefftz-plane value exactly on straight wings, but on the TU Delft V3 kite it is 7–15 % low and decreases without bound as the number of spanwise panels increases. This document states the problem, identifies its cause (the logarithmically singular self-induction of the curved bound vortex, evaluated on its own line to set the force direction, entering the drag through $\rho\Gamma\, u \sin\alpha$), reviews how the literature treats the singularity, tests the two published remedies inside the VSM framework, and describes the parameter-free regularisation now implemented: the quarter-chord velocity is built from the classical lifting-line downwash of the wing unrolled onto the local tangent line, plus the difference between the actual and the unrolled vortex systems with the bound vorticity and the leg starting points spread chordwise with the thin-airfoil distribution. The result is identically zero on straight wings, mesh-independent, and within 1 % of the Trefftz-plane induced drag on an arc wing, a swept wing, the swept–anhedral elliptic wing of Gaunaa et al., and the V3 kite. A second part settles the direction of the attached-trailed legs, which the theory leaves open, against a vortex-lattice referee: the paper's perpendicular legs lose 8–9 % of the lift on kinked and kite-like wings, legs along the ribs or along the flow reproduce the lifting-surface lift within 1 %. All tables are reproduced by the scripts in `docs/validation`.

1 Status of the implementation with respect to Li and Gaunaa

Since the September 2026 commits (`233a238`…`62c53af`) the code implements the following elements of Gaunaa, Li and Pirrung [1] and Li, Gaunaa, Pirrung and Lønbæk [3]:

- CP1: the inner 2D velocity is the component perpendicular to the local quarter-chord line, and the chord is measured perpendicular to that line;
- TAT1: the section's own 2D bound-vortex induction is subtracted at the three-quarter-chord point (the VSM `subtract_bound_2D` term, which the code has had since its inception);
- TAT2: force magnitudes from the three-quarter-chord angle of attack, with $\Gamma_i = \frac{1}{2}|V_{\mathrm{inner},i}|\, c_i\, C_l(\alpha_{3/4,i})$ (eq. 4 of [1]);
- TAT3: force directions from the undisturbed relative flow at the quarter chord (`is_aoa_corrected=True`, now the default);
- the attached-trailed (AT) vortex force $\rho\,\Delta\Gamma_j\,\mathbf{V}_{3D,\mathrm{rel},j} \times \mathbf{l}_{AT,j}$ evaluated at the three-quarter-chord point of the leg;
- the implicit artificial viscosity with the tip closures and the relaxation-factor bound of [3], the latter halved for the VSM after measurement of the fixed-point Jacobian.

Two elements do not follow the published prescription.

1. *The curved bound-vortex influence on the direction velocity.* Reference [1] states that LL-Gaunaa includes "the influence of the curved bound vortex" according to [5]. In [5] (Section 2.2) that influence is the induction of the 3D bound vortex minus that of an infinitely long 2D vortex of the section's own strength, evaluated at the *three-quarter* chord, assumed constant along the chord, and added to the quarter-chord velocity. `develop` instead evaluated the other panels' bound segments *on the quarter-chord line itself* (commit `9c43c1a`), which [5] calls LL-P&S and shows to be log-singular and non-convergent. This is the cause of the problem treated here (Section 2). A literal implementation of the [5] recipe in this code was also tested and fails on straight wings (Section 3); the form that reproduces the authors' published behaviour, curved-minus-straight bound segments at the control point, works with their perpendicular legs and not with rib legs (Table 2), and needs confirmation with them. The branch replaces the term by the regularisation of Section 4.
2. *The direction of the attached-trailed legs.* Reference [1] (Section 3) states that the framework "does not indicate how the trailed vorticity should connect from the 1/4 chord junctions to the trailing edge", that this "is not specified by theoretical results" and "is effectively a modeling choice", and aligns the legs "with the local chordwise direction up to the trailing edge", i.e. perpendicular to the quarter-chord line (CP1). The code kept the legs along the input ribs; the paper's construction and a streamwise one were implemented (`leg_direction`) and the three were compared against a vortex-lattice referee (Section 4.4, Figures 5 and 6). The perpendicular legs lose 9 % of the lift on a kinked swept wing and 8 % on the V3, where they also cross at the tips above 80 panels; rib and streamwise legs reproduce the lifting-surface lift within 1 % on straight, yawed, kinked, arc and swept–anhedral elliptic wings (streamwise: 2 % low on the arc). The rib legs remain the default (Section 4.4, Decision); the streamwise construction is available as an option. This is now a documented choice backed by a referee rather than an untested deviation.

2 Problem statement

2.1 Observation

For steady, uniform inflow the induced drag can be computed in two ways from the same converged circulation $\Gamma(s)$: on the blade, by projecting the panel forces (lift tilted by the quarter-chord flow angle, plus the AT force) onto the freestream, or in the Trefftz plane from the momentum in the far wake. The Trefftz value depends only on $\Gamma(s)$ and the wake geometry; for a consistent force evaluation the two must agree. Table 1 shows the ratio for inviscid polars ($C_l = 2\pi\alpha$) with the bound segments evaluated on the line.

Table 1: On-blade over Trefftz-plane induced drag, bound segments on the quarter-chord line (`develop` as of `62c53af`), inviscid polars.

case	$N = 10$	20	40	80	160	trend
straight rectangular, AR 8	–	1.000	1.000	1.000	–	exact
swept 30°, AR 8, streamwise ribs	0.933	0.937	0.939	0.940	0.936	converged, 6 % low
circular arc, 110°, $R = 4c$	0.998	0.977	0.957	0.937	0.917	−0.02 per doubling
elliptic, 10 % sweep, 25 % anhedral ([1])	–	–	0.980	0.967	0.954	−0.013 per doubling
V3 kite, CAD geometry	–	0.933	0.907	0.881	0.854	−0.026 per doubling

Lift is converged to four digits in all cases; only the drag drifts. The drift is a clean logarithm in N on every wing with out-of-plane curvature.

2.2 Cause

The quarter-chord direction velocity of panel i was assembled as

$$\mathbf{V}_{1/4,i} = \mathbf{V}_\infty + \sum_j \Gamma_j \, \mathbf{K}_j^{\text{trailed}}(\mathbf{P}_i) + \sum_{j \neq i} \Gamma_j \, \mathbf{K}_j^{\text{bound}}(\mathbf{P}_i), \tag{1}$$

with $\mathbf{P}_i$ on the bound line. The last sum is needed: without it a swept wing with streamwise legs diverges (ratio 0.43 at $N = 40$, 0.25 at $N = 160$), because the legs carry a logarithmic term of their own that only the bound segments cancel. But on a line curved out of its own plane the neighbouring segments induce, at $\mathbf{P}_i$, a velocity perpendicular to the plane of curvature, i.e. along the chord,

$$u \simeq \frac{\Gamma\kappa}{4\pi} \ln \frac{L}{\epsilon}, \qquad \epsilon \sim \frac{\text{panel width}}{2}, \tag{2}$$

which has no counterpart in the legs and grows as the panels shrink. Figure 1(b) shows this on the arc: the polyline value at mid-span grows by 0.05 m/s per doubling of N ($U = 10$ m/s).

The drag is exceptionally sensitive to this component. With the lift perpendicular to $\mathbf{V}_{1/4}$ and $\mathbf{V}_{1/4} = (U\cos\alpha + u,\ U\sin\alpha - w)$ in chord/normal components, the freestream component of the force is

$$D_i = \rho \, \Gamma_i \, (w\cos\alpha + u\sin\alpha). \tag{3}$$

The downwash w corresponds to about one degree of induced angle, so a chordwise u of 3 % of U at $\alpha = 8°$ moves the induced drag by about 12 %. This is the whole V3 shortfall: on the arc the ratio is 1.08 with the bound term removed and drifts from 0.96 to 0.92 with it included; the physically correct value of u lies in between and is finite.

3 How the literature treats the singularity

3.1 Li, Gaunaa, Pirrung, Ramos-García, Horcas (2020): evaluate at the three-quarter chord

Reference [5] identifies the LL-P&S evaluation (bound vortex on its own line, as in Phillips and Snyder [11]) as log-singular and demonstrates its mesh dependence numerically (their Figs. 3–4). Their remedy (LL-mod) evaluates the bound-vortex induction at the three-quarter-chord point, subtracts the 2D infinite-vortex induction of the section's own strength, treats the difference as constant along the chord and adds it to the quarter-chord velocity that sets the force direction. The trailed vortices are also evaluated at the three-quarter chord in LL-mod, which they note makes the iteration less stable. This is the reference cited by [1] for LL-Gaunaa; neither the 2026 paper nor its online appendix [2] (results for ten planforms at 120 sections, no convergence study) adds implementation detail.

Test in the VSM framework. The recipe was implemented literally: for every panel j, the finite bound segment at the control point $\mathbf{C}_i$ minus (for $j = i$) the infinite line, reused at $\mathbf{P}_i$. Results (on-blade/Trefftz): straight AR 8 wing 0.848, straight AR 4 wing 0.815, arc 0.84, swept–anhedral elliptic wing with perpendicular legs 0.865, swept 30° wing 0.50 → 0.35. The straight-wing failure is decisive: for a finite wing with spanwise-varying Γ the "3D minus 2D" difference at the three-quarter chord is not zero (it contains the finite-span tails and the Γ-variation of the bound vortex; +0.02 m/s at mid-span and +0.11 m/s near the tips at $U = 20$ m/s on the AR 8 wing), and eq. (3) amplifies it. The same quantity used for the three-quarter-chord angle of attack, which the VSM has always done, is correct and untouched; only its reuse as a quarter-chord velocity fails. Since [1] reports LL-Gaunaa within 0.6 % of a VLM on straight wings, the authors' implementation must differ from this literal reading, most plausibly by taking "curved minus straight" (as first described in [6]) rather than "3D minus 2D". This should be confirmed with the authors. In their appendix the on-blade

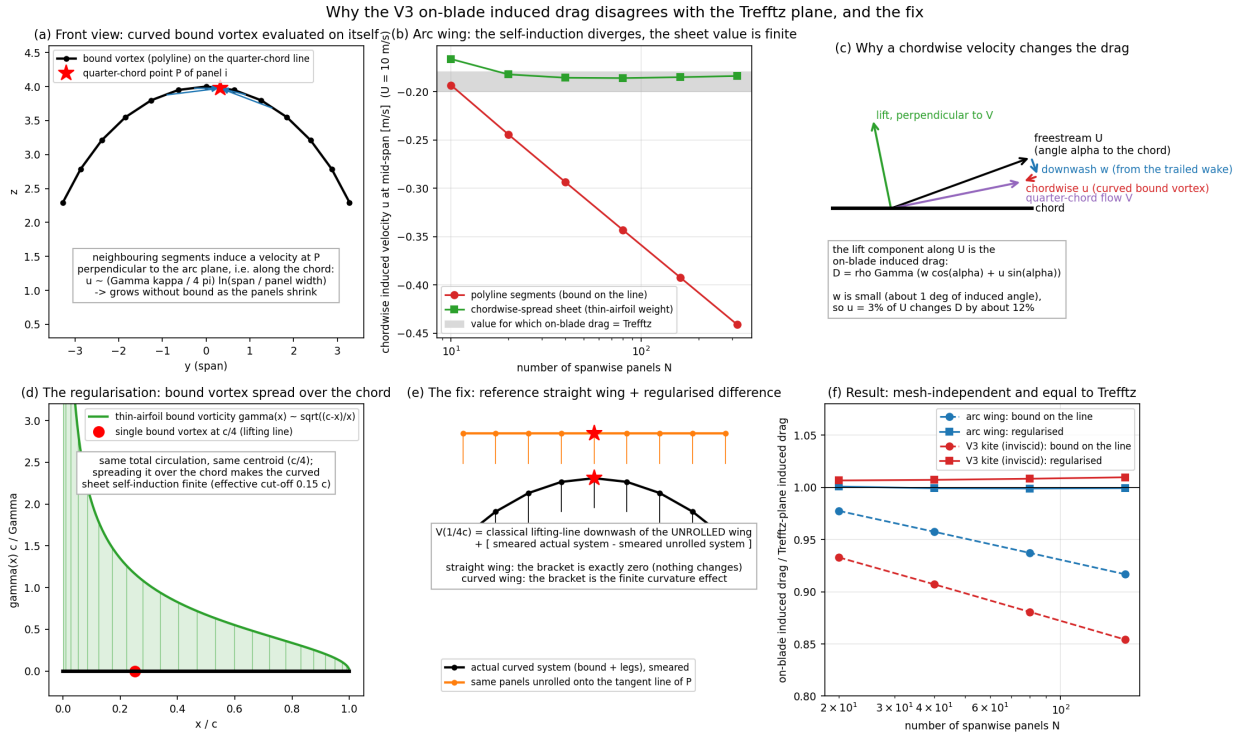


Figure 1: Mechanism and fix (`docs/figures/08`). (a) Front view of the curved bound polyline with the quarter-chord point of panel i. (b) Arc wing, mid-span: chordwise self-induction from the polyline (grows logarithmically with N) and from the chordwise-spread sheet (finite); grey band: value at which on-blade drag equals Trefftz. (c) Why a chordwise velocity changes the drag, eq. (3). (d) The thin-airfoil bound vorticity distribution used for the regularisation; effective cut-off $\exp\langle \ln|x - c/4| \rangle_\gamma = 0.15\,c$. (e) The unrolled reference construction. (f) Convergence of on-blade over Trefftz drag before and after.

Oswald factor sits 5.6 % and 1.1 % above the Trefftz one for the 15 % and 25 % swept wings, within 0.6 % for the dihedral, anhedral and combined wings.

The hypothesised form, tested. The reading that fits all their published numbers is: the bound segments of the real wing minus those of the wing unrolled onto the section's tangent, single filaments, evaluated at the three-quarter-chord control point where every distance is at least half a chord, and reused at the quarter chord (`_curved_minus_straight_cp` in the validation script, `--hypothesis`). It is zero on straight wings by construction and finite everywhere. Table 2 shows what it does here. With the paper's perpendicular legs it is mesh-independent and within 1–2 % of the Trefftz plane on the elliptic wing and the V3, i.e. it is a working regularisation of the bound vortex, with the half-chord distance to the control point as its cut-off (on the arc it gives 0.993 against 0.999 for the sheet's 0.15 c). With the legs along the ribs it fails on the V3, 0.81 falling to 0.74, because legs that are not perpendicular to the bound line carry a logarithm of their own at the quarter-chord point (the same one that makes the swept wing with streamwise legs diverge in Table 1 without the bound term), and a bound-only difference at the control point cannot see it. The authors never meet that case because CP1 makes their legs perpendicular; a kite with ribs does. This is the most likely reason their code converges where the literal reading of [5] does not, and the reason the construction of Section 4 differences the whole vortex system, legs included, rather than the bound segments alone.

Table 2: The hypothesised LL-Gaunaa quarter-chord term (curved minus straight bound segments, single filaments, at the control point, reused at the quarter chord): on-blade over Trefftz induced drag, inviscid polars, with rib legs and with the paper's perpendicular legs.

case	legs	$N = 20/40$	80	160	trend
straight AR 8	either	1.000	1.000	–	exact
circular arc, 110°, $R = 4c$	either	0.992 / 0.993	0.993	0.993	converged, 0.7 % low
elliptic, swept–anhedral	rib	0.993	0.986	0.979	drifting
elliptic, swept–anhedral	perpendicular	1.020	1.017	1.012	converging
V3 kite	rib	0.809	0.772	0.735	diverging
V3 kite	perpendicular	1.006	1.015	–	converged
swept 30°, kinked	rib	0.908	0.840	–	diverging
swept 30°, kinked	perpendicular	1.129	1.138	–	fan of legs at the kink

3.2 Reid and Hunsaker (2020), Goates and Hunsaker (2021, 2023): effective lifting line and jointed legs

References [7, 8, 9] (open-source implementation MachUpX) prove that the induced velocity on a lifting line is finite only where the line's second derivative vanishes and the trailing vortices leave it perpendicularly. Their construction: each control point sees an *effective* lifting line, straight (tangent) near the point and blended into the true curve with a Gaussian of prescribed width ("blending distance", recommended $\gtrsim 25$ % of the span, MachUpX default one semispan); each trailing vortex has a finite *joint* of length δc perpendicular to the effective line before turning along the freestream (recommended $\delta \gtrsim 0.15$, MachUpX default 0.15). Bound segments of the other sections are kept. Validation is on circulation distributions and C_L of planar swept wings against panel methods and experiment; induced drag and non-planar wings are not examined.

Test in the VSM framework. The construction was implemented for the quarter-chord direction velocity, everything else unchanged (Table 3). It is mesh-independent, as claimed, but the induced drag depends on the two lengths, by 17 % on the V3 and 12 % on the arc across reasonable choices, because the blending distance is precisely the cut-off of the curvature logarithm; the authors' insensitivity statement concerns lift, which is far less sensitive. With streamwise rib legs the swept wing fails (0.2–0.7), since the perpendicular joints no longer represent the vortex system that carries the force.

Table 3: Reid–Hunsaker construction in the VSM framework: on-blade over Trefftz drag at $N = 80$. Joint length and blending distance in local chords, except the MachUpX column (defaults: joint $0.15\,c$, blending one semispan). Last column: the regularisation of Section 4 at the same N.

case	0.25 / 1	0.5 / 1	1.0 / 1	0.5 / 0.5	0.5 / 2	MachUpX	regularised
straight AR 8	1.000	1.000	1.000	1.000	1.000	1.001	1.000
swept 30°, rib legs	0.642	0.471	0.198	0.191	0.637	0.688	0.982
arc, $R = 4c$	1.049	1.025	1.004	1.003	1.052	1.117	0.999
elliptic, 10 % sweep, 25 % anhedral	1.007	0.988	0.968	0.968	1.008	1.042	1.015
V3 kite	1.003	0.966	0.930	0.885	1.032	1.053	1.008

3.3 Other approaches

- *Li et al. 2018* [6]: correct a straight-blade model by the difference between the curved and an imaginary straight bound vortex. Same subtraction idea as below, but with single filaments, hence mesh dependent; abandoned in [5].
- *Guermond 1990* [10]: matched-asymptotic lifting-line theory for curved and swept wings. Shows analytically that the curvature term is finite once the inner (airfoil-scale) problem is accounted for, using Hadamard finite-part integrals. It is the theoretical basis for a chord-scale cut-off but is not a numerical scheme with horseshoes and polars.
- *Each segment minus its own infinite line, at the quarter chord.* The natural extension of TAT1 to the neighbours: for every panel j subtract from its finite bound segment the infinite line through it. It works for the own section at the three-quarter chord, where the point is half a chord from the line, and fails at the quarter-chord point, which lies at a distance from a neighbour's line that shrinks with the square of the arc-length separation: the infinite-line term then grows like one over the panel width squared. On the arc wing the mid-span sum goes -27, -110, -448, -1805 at 20, 40, 80, 160 panels, against $+0.075$ to $+0.117$ for the segments alone. Tested here, not usable.
- *Vortex core on the bound vortex* (e.g. QBlade [13], core radius a fraction of chord): Reid and Hunsaker note these are "very sensitive to the cutoff/core distance". Tested here with the code's Vatistas kernel: the projection of an on-line point to the core boundary produces spurious velocities of order $\Gamma/(4\pi\epsilon)$ and unusable results.
- *Three-quarter-chord directions* (Weissinger, Ranneberg [12], the VSM default before the `develop` branch): no singularity, but the induced drag is overestimated by 20–40 % on the V3 [1].
- *Trefftz-plane drag only* (Ranneberg): exact for the total in steady uniform inflow, unavailable with body rates or distributed inflow, no spanwise distribution.

4 The regularisation implemented on the branch

4.1 Formulation

Let $\mathbf{K}_j^{\mathrm{LL}}(\mathbf{P})$ denote the induction at $\mathbf{P}$ of ring j without its bound segment (two legs and two semi-infinite wake filaments; this is what the code's LLT influence matrix holds), and let $\mathbf{K}_j^{\mathrm{sheet}}(\mathbf{P})$ denote the induction of ring j with its bound vorticity and its leg starting points spread along the chord:

$$\mathbf{K}_j^{\mathrm{sheet}}(\mathbf{P}) = \sum_{q=1}^{M} w_q \Big[\mathbf{K}^{\mathrm{seg}}\big(\mathbf{n}_{2,jq} \to \mathbf{n}_{1,jq}; \mathbf{P}\big) + \mathbf{K}^{\mathrm{leg}}\big(\mathbf{n}_{1,jq} \to \mathbf{T}_{1,j}\big) + \mathbf{K}^{\mathrm{leg}}\big(\mathbf{T}_{2,j} \to \mathbf{n}_{2,jq}\big) \Big] + \mathbf{K}_j^{\mathrm{wake}}(\mathbf{P}), \tag{4}$$

where the strip nodes lie on each node's own leg line,

$$\mathbf{n}_{k,jq} = \mathbf{b}_{k,j} + \frac{x_q - x_{ac}}{1 - x_{ac}} \, (\mathbf{T}_{k,j} - \mathbf{b}_{k,j}), \qquad x_{ac} = \tfrac{1}{4}, \tag{5}$$

so that the strip legs are collinear with the actual legs, and the chordwise stations and weights follow the thin-airfoil distribution $\gamma(x) \propto \sqrt{(c-x)/x}$ through $x_q = \sin^2\phi_q$, $w_q \propto \cos^2\phi_q$, $\phi_q = (q-\frac{1}{2})\pi/(2M)$, $M = 24$ (`thin_airfoil_strips`). The distribution has its centroid at $c/4$, so the far field of the sheet equals that of the single filament. Figure 2 shows the strips on real and unrolled panels and what the spreading does to the induction of a neighbouring segment: nothing beyond a chord, a finite value instead of a growing one within it.

For each evaluation panel i, the *unrolled* reference wing is built on the tangent line $\mathbf{t}_i$ of panel i through $\mathbf{P}_i$: every node keeps its arc-length coordinate along the bound line, the legs are perpendicular to the line in panel i's chord direction, each with its own panel's chord length, and the wake keeps its direction. There is one such reference wing per evaluation panel. The quarter-chord velocity is then

$$\boxed{\mathbf{V}_{1/4,i} = \mathbf{V}_\infty + \sum_j \Gamma_j \Big[\mathbf{K}^{\mathrm{LL}}_{j,\mathrm{unrolled}(i)}(\mathbf{P}_i) + \mathbf{K}^{\mathrm{sheet}}_{j}(\mathbf{P}_i) - \mathbf{K}^{\mathrm{sheet}}_{j,\mathrm{unrolled}(i)}(\mathbf{P}_i)\Big].} \tag{6}$$

Figure 4 shows the construction. The first term is Prandtl's classical downwash of the straight wing with the actual $\Gamma(s)$, exact for straight wings (ratio 1.000). The bracket is the curvature, sweep and skew effect; it is identically zero on a straight wing with perpendicular legs, finite because both sheets regularise the near field at the chord scale, and convergent because no filament is evaluated on its own line. The semi-infinite wake filaments cancel identically between the actual and unrolled systems, so the wake model plays no role. Nothing else changes: the three-quarter-chord magnitude (which already contains the LL-mod term of [5]), the circulation map, the AT force and the wake are untouched. The single-filament limit of the bracket recovers the curved-minus-straight idea of [6]; the chordwise spreading is the discrete counterpart of the inner solution in [10]; the local straightening is the mechanism of [7] without its two empirical lengths. The effective cut-off implied by the distribution is $\exp\langle \ln|x - c/4|\rangle_\gamma = 0.1515\,c$; the number of strips does not matter beyond $M \approx 20$ (24 and 32 strips give the same arc self-induction to three digits). Implementation: `assemble_curved_bound_correction_AIC` in `src/VSM/core/utils.py`, called from `BodyAerodynamics.update_effective_angle_of_attack_if_VSM`; about 0.5 s at 50 panels and 1 s at 80. The matrix depends on the panel geometry only, so it is computed once per shape and reused for every inflow and circulation; a body that is the cached shape up to a rigid rotation and translation (a trim that rotates the kite for every residual evaluation) reuses the stored matrix rotated, and only a deformation recomputes it.

4.2 Reading of the formulation

Three remarks make eq. (6) less opaque than it looks.

The rule is the one the VSM already uses at the three-quarter chord. The outer problem is the inviscid vortex system, the inner problem the 2D section with its polar, and the coupling hands the inner model the flow it does not already contain. A 2D section is an infinite straight wing: an infinitely long straight bound sheet along the chord with a straight wake. At the three-quarter chord the own infinite bound vortex is subtracted (TAT1) because the polar has it; on a straight wing the neighbours are collinear with that infinite vortex and add nothing, and the whole trailed system stays in because the 2D model has none of it. At the quarter chord the same subtraction is made: the own straight continuation is the unrolled wing, the section's tangent line with the neighbours laid on it, so "real minus unrolled" is "3D minus what the profile already knows about itself and its straight continuation", applied to the whole system. On a straight wing the two coincide and nothing is handed over beyond Prandtl's downwash.

The sheet is what the profile is made of, not a numerical device. The three-quarter-chord sums converge (Table 4): the control point is half a chord from every filament, the lumped vortex at the quarter chord with the control point at the three-quarter chord reproduces thin-airfoil lift exactly (Pistolesi), and a sheet there would change that equivalence without regularising anything. The quarter-chord point lies on the line, and there the neighbours sit at a fraction of a chord. What is

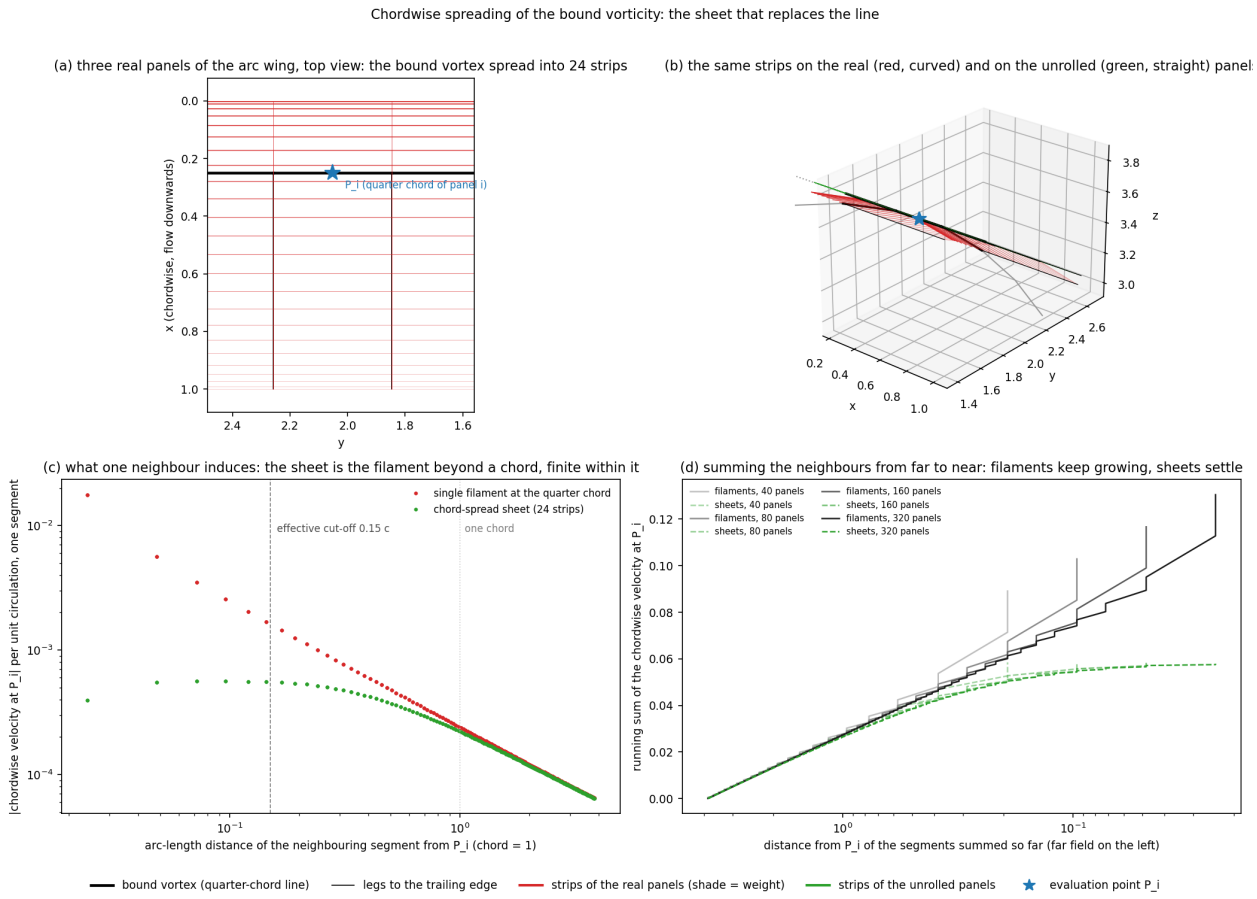


Figure 2: The chordwise spreading (`docs/validation/chord_spread_figure.py`). (a) Three panels of the arc wing from above: the bound vortex of each panel replaced by 24 strips at the thin-airfoil stations, line weight and shade proportional to the strip weight, each strip with its own legs on the node's leg line. (b) The same strips on the real, curved panels (red) and on the unrolled, straight panels of the middle panel's reference (green): in the panel's neighbourhood the two sheets nearly coincide, which is why their difference is small there. (c) The chordwise velocity that one neighbouring bound segment of unit strength induces at the quarter-chord point $\mathbf{P}_i$ against its arc-length distance, as a single filament and as the spread sheet: beyond a chord the two are the same, within a chord the filament grows without bound and the sheet levels off at the cut-off $0.15\,c$ that the distribution fixes. (d) The running sum of those contributions from the far field inwards for 40 to 320 panels: with filaments every refinement adds a new nearest neighbour and the sum keeps rising, the logarithm of Li et al.; with sheets the sum settles to the same value at every panel count.

there, in the inner model's own terms, is the neighbour's chordwise sheet, bent slightly out of the section's plane; two sheets at a small angle induce a finite velocity on each other, two lines do not. Spreading the bound vorticity with the thin-airfoil distribution therefore uses, for the neighbours, the vorticity the inner model already uses for the section itself; the cut-off $0.15\,c$ is that distribution's, not a choice. The unrolled system is spread with the same strips so that collinear neighbours, which are the own 2D vortex, cancel exactly. The first term of eq. (6) is left as single filaments because it is Prandtl's exact downwash of the straight wing, which spreading the leg starts would alter; the spreading enters only through the difference, which is zero wherever the wing is straight.

Why two lines diverge and two sheets do not (Figure 3). Let line 2 start where line 1 ends, turned by an angle θ out of their common plane, and let $\mathbf{P}$ sit on line 1 at distance r from the kink. The velocity line 2 induces at $\mathbf{P}$, perpendicular to the plane of the two lines and hence along the chord, is

$$v = \frac{\Gamma\,(1-\cos\theta)}{4\pi\, r \sin\theta} \;\approx\; \frac{\Gamma\theta}{8\pi r}, \tag{7}$$

unbounded as $\mathbf{P}$ approaches the kink for any $\theta \neq 0$. A smooth line of curvature κ is a continuum of such kinks: the element at arc distance d is seen under the angle $\kappa d/2$ and contributes $\mathrm{d}v = \Gamma\kappa\,\mathrm{d}s/(8\pi d)$, and the sum from the nearest element at the panel width Δs to the span L is $(\Gamma\kappa/8\pi)\ln(L/\Delta s)$ per side: the logarithm, one constant per doubling, no limit because the nearest element is always at the panel width. Give each line its chordwise sheet and let $\mathbf{P}$ sit at chord station x_P: the element of sheet 2 at station x and arc distance d is at distance $\sqrt{d^2+\Delta x^2}$, $\Delta x = x - x_P$, and the same geometry gives

$$\mathrm{d}v = \frac{\Gamma\gamma(x)\,\mathrm{d}x}{4\pi}\;\frac{\kappa d^2/2}{(d^2+\Delta x^2)^{3/2}}\,\mathrm{d}s, \tag{8}$$

which is the line's $\Gamma\kappa\,\mathrm{d}s/(8\pi d)$ for $d \gg |\Delta x|$ and vanishes like d^2 for $d \ll |\Delta x|$: seen from more than a chord away the sheet is the line, seen from within the chord the chordwise offset keeps the element away from $\mathbf{P}$ (at the kink itself the induced velocity is perpendicular to the purely chordwise vector from the element to $\mathbf{P}$ and has no chordwise component at all). Integrated over d the element gives $(\Gamma\kappa\gamma\,\mathrm{d}x/8\pi)[\ln(2L/|\Delta x|)-1]$, finite, with the chordwise offset in the place of the panel width; averaged over the chord with the thin-airfoil weight, $\langle\ln|x-x_P|\rangle_\gamma$ is finite because the logarithm is integrable, and the result is $(\Gamma\kappa/8\pi)\ln(L/c_{\mathrm{eff}})$ with $c_{\mathrm{eff}} = \exp\langle\ln|x-c/4|\rangle_\gamma = 0.15\,c$. The mesh has dropped out. In physical terms a line vortex is a singularity and two singularities touching at an angle have no finite mutual velocity; a sheet is a regular distribution, the 2D airfoil induces the finite, uniform $\Gamma/(\pi c)$ on its own chord, and two sheets touching along a rib see a finite velocity. The neighbouring sections of a curved wing are sheets touching along their common rib.

The two sheet terms are the same panel in two positions. $\mathbf{K}^{\mathrm{sheet}}_{j}$ is panel j where it is, on the curved line, with its legs along its rib; $\mathbf{K}^{\mathrm{sheet}}_{j,\mathrm{unrolled}(i)}$ is the same panel with the same strips laid on panel i's tangent at the same arc-length distance, with its leg in panel i's chord direction. Biot–Savart depends on where a segment is and how it is oriented with respect to $\mathbf{P}_i$, so the two differ wherever the wing is curved: by little for the neighbours, whose two positions almost coincide (Figure 4d), and by the whole curvature effect for the far panels. Nothing singular is ever subtracted from anything singular.

Table 4: Why the three-quarter chord needs no sheet. Arc wing, mid-span panel, unit circulation on every other panel, bound segments as single filaments: the normal component at the control point converges (the neighbours build up the infinite-line value $1/(2\pi\cdot c/2) = 0.318$ that TAT1 subtracts); the chordwise component at the quarter-chord point adds a constant per doubling.

panels	20	40	80	160	320
at the 3/4-chord control point, normal	0.198	0.252	0.282	0.297	0.305
at the 1/4-chord point, chordwise	0.075	0.089	0.103	0.117	0.130

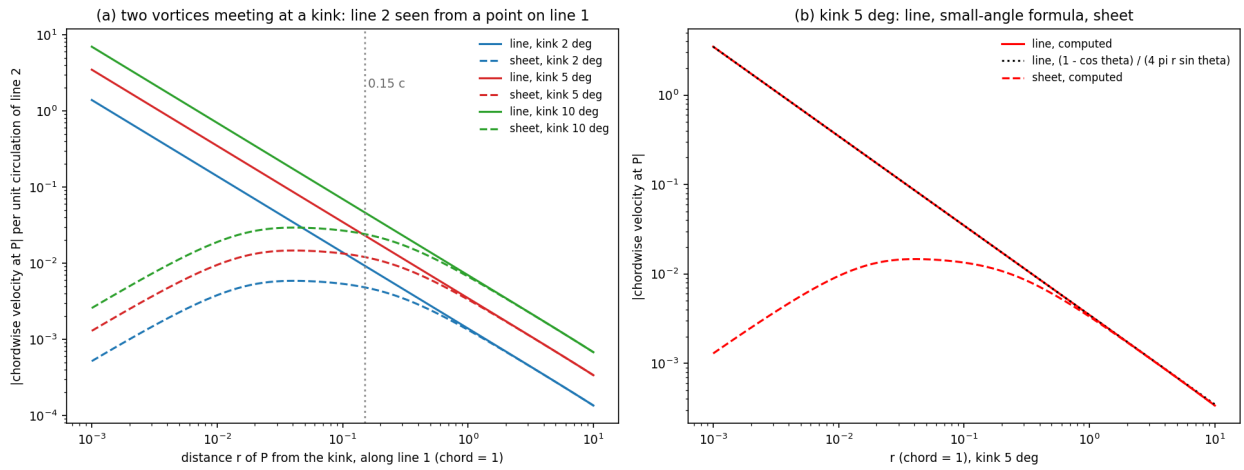


Figure 3: Two vortices meeting at a kink of angle θ (`docs/validation/kink_figure.py`): the chordwise velocity that line 2, of unit strength, induces at a point of line 1 at distance r from the kink, as a single filament and as the chord-spread sheet. (a) Kinks of 2, 5 and 10°: the filament follows eq. (7), a line of slope -1 without limit; the sheet coincides with it beyond about a third of a chord, is bounded within, and vanishes at the kink. (b) The 5° case with the small-angle formula.

4.3 Verification against the Trefftz plane

Table 5 repeats Table 1 with eq. (6) in place: the ratio is 1.000 on the straight wing, within 1 % on the arc and the V3 at every panel count, and no longer drifts.

Table 5: On-blade over Trefftz-plane induced drag with eq. (6), inviscid polars (`curved_bound_vortex_validation.py`).

case	$N = 10$	20	40	80	160
straight rectangular, AR 8	–	–	1.000	1.000	–
straight, ribs skewed 22°	–	–	1.029	1.030	–
swept 30°, streamwise ribs	–	–	0.971	0.982	0.988
circular arc, 110°, $R = 4c$	1.008	1.001	0.999	0.999	0.999
elliptic, 10 % sweep, 25 % anhedral, rib legs	–	–	1.014	1.015	1.017
elliptic, same, perpendicular legs	–	–	1.025	1.024	1.020
V3 kite, rib legs	–	1.011	1.007	1.008	1.010
V3 kite, perpendicular legs	–	–	1.015	1.018	crosses

Two residuals remain and are unrelated to the bound vortex: the swept wing converges to 0.99 rather than 1.00, and the straight wing with skewed ribs sits at 1.03 with or without the fix. Both point to the attached-trailed force on legs that are not aligned with the flow and are a separate, smaller item, taken up in Section 4.4. The perpendicular-leg rows use the construction of Section 4.4; on the V3 it cannot be built above 80 panels.

4.4 Direction of the attached-trailed legs

Theory fixes the inner 2D section (CP1: velocity, chord and angle of attack perpendicular to the quarter-chord line) but not the path of the attached-trailed vorticity from the bound vortex to the trailing edge; reference [1] calls it a modelling choice and takes the CP1 chord direction. Three constructions are implemented (`BodyAerodynamics(leg_direction=...)`, Figure 5):

- `rib` (default): along the interpolated input sections to their trailing-edge points, control point on the section at the control-point chord fraction; a fixed vortex system, independent of the

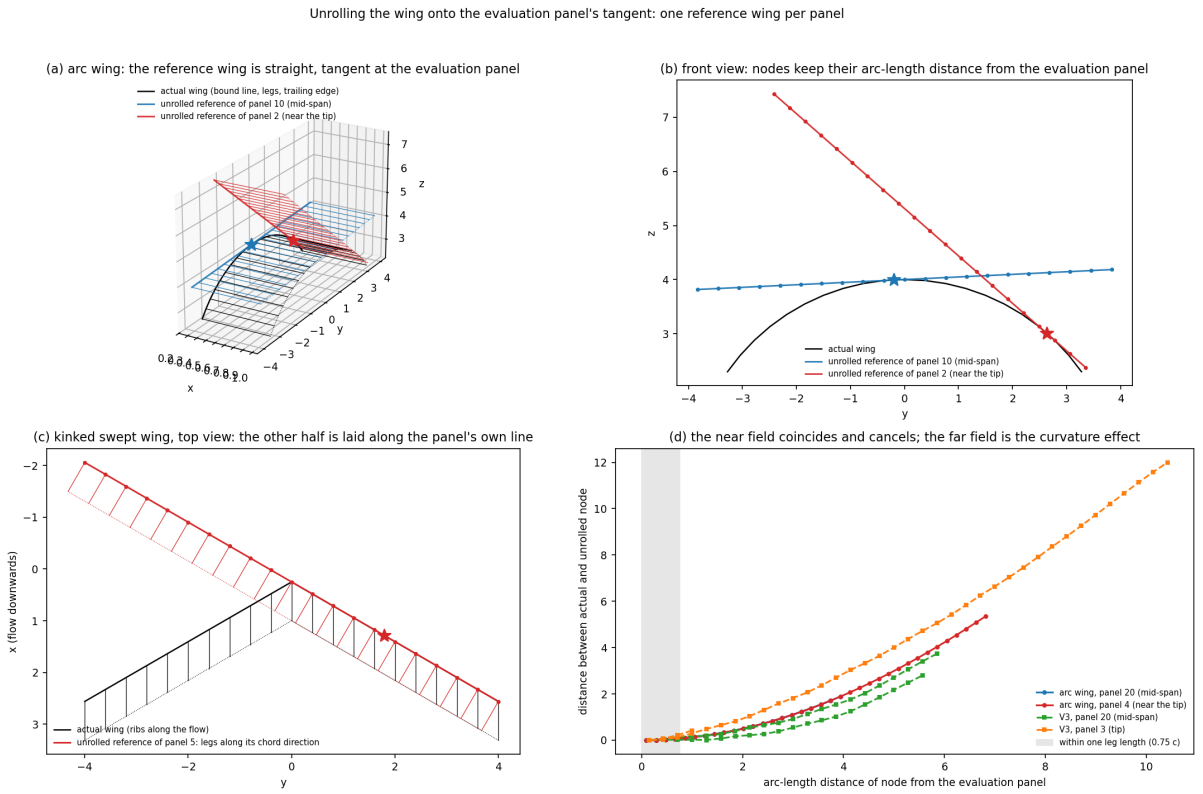


Figure 4: What "unrolling" means (`docs/validation/unrolling_figure.py`). For an evaluation panel i (star) the reference wing is built on panel i's own tangent line through its aerodynamic centre: every node keeps its arc-length distance from the panel, the legs are perpendicular to the line in panel i's chord direction with each panel's own chord, and the wake keeps its direction. There is one such reference wing per panel. (a, b) The arc wing with the references of a mid-span panel (blue) and a panel near the tip (red): the two straight wings are different, each tangent at its own panel. (c) The kinked swept wing seen from above, with the reference of a panel on the right half: the whole wing, including the other half, is laid along that panel's line, and the legs follow that panel's chord direction. (d) The distance between the actual and the unrolled nodes against their arc-length distance from the evaluation panel, for the arc wing and the V3: within a leg length the two systems coincide to second order, so the singular near field cancels in the difference of eq. (6); what remains is the far-field effect of the curvature, finite once both systems are spread over the chord.

inflow. On a kite these are the ribs of the real wing; on the test wings they are drawn along the flow (swept wings), along the chord (arc) or deliberately skewed (straight wing).
- `streamwise`: along the mean apparent velocity projected onto the local surface, up to the trailing-edge curve, rebuilt whenever the inflow is set; a Weissinger-type lattice. With a spanwise-varying inflow (body rates, distributed inflow) only the mean direction is used.
- `perpendicular`: the paper's choice. The tangent of the quarter-chord line is averaged over plus and minus one leg length, $(1-a)c$, before the perpendicular is taken, so that the legs fan out instead of crossing at a planform kink (Figure 5b); the averaging length is the leg's own scale, not a free parameter. Legs that still cross raise an error.

In the last two the leg ends at the point of the trailing-edge curve closest to the leg's ray (a ray does not meet a polyline on a curved surface exactly), the curve being extended beyond its ends so that a leg leaving a skewed tip still ends on the trailing-edge line, and the control point moves with the legs so that it stays between the panel's own two legs (midway between parallel legs this is the paper's 3/4-chord point of the CP1 section; without this move the control points of a kinked wing land on neighbouring legs and the solve diverges). The curved bound-vortex regularisation of Section 4 and the AT force follow whichever legs are built.

Referee. The Trefftz plane checks the induced drag of a given circulation but cannot judge the lift. The vortex-lattice method of AeroSandbox [14] was therefore run on the same wings with flat-plate sections ($C_l = 2\pi\alpha$, as in the VSM runs here), 16 chordwise by 80–160 spanwise panels, trailing legs along the body axis (`vlm_referee.py`). For the non-planar wings its near-field drag is 8–14 % below the Trefftz drag of its own circulation, so the latter is used as the reference. For the V3 the referee is approximate: AeroSandbox takes its sections perpendicular to the local span, which unskews the tip ribs and moves the trailing edge there. Figure 6 and Table 6 give the result; Table 7 the on-blade over Trefftz ratios and the V3 polars against RANS.

Table 6: Lift and induced drag of the VSM with the three leg options relative to the vortex-lattice referee (finest meshes; VSM at $N = 80$, elliptic $N = 160$). Induced drag: VSM on-blade over VLM Trefftz.

	VLM		rib		streamwise		perpendicular	
case	C_L	C_{D_i}	C_L	C_{D_i}	C_L	C_{D_i}	C_L	C_{D_i}
straight AR 8, 5°	0.402	0.00654	−0.2 %	−0.9 %	−0.2 %	−0.9 %	−0.2 %	−0.9 %
yawed 30°, 5°	0.360	0.00537	0.0 %	−0.6 %	0.0 %	−0.6 %	+1.4 %	+2.0 %
kinked swept 30°, 5°	0.363	0.00568	−0.4 %	−3.0 %	−0.4 %	−3.0 %	−8.8 %	+6.7 %
arc 110°, 8°	0.577	0.01481	+0.3 %	−3.4 %	−2.1 %	−1.2 %	+0.3 %	−3.4 %
elliptic, swept–anhedral, 10°	0.807	0.02643	+0.9 %	−0.1 %	+0.1 %	−4.4 %	+0.8 %	−0.3 %
V3, flat plate, 10° (approx.)	0.624	0.02961	+2.3 %	−4.0 %	−1.9 %	−8.6 %	−7.9 %	−4.5 %

What the referee says. On the planar wings the streamwise legs, which coincide with the ribs there, reproduce the lifting-surface lift to 0.4 % and the induced drag within 3 %; the perpendicular legs are 1.4 % high on the yawed wing and 8.8 % low on the kinked wing, with the induced drag 6.7 % high. A perpendicular lattice is a sheared Weissinger lattice; at a kink the fan of legs changes the downwash at the neighbouring control points, and even without a kink the sheared lattice is not the better discretisation. On the arc the rib and perpendicular legs coincide (chordwise) and are within 0.3 % in lift; the streamwise legs, which climb along the vertical tips, are 2.1 % low. On the elliptic wing all three are within 1 % in lift, the streamwise legs closest. On the V3 the approximate referee puts the rib legs 2.3 % high and the streamwise legs 1.9 % low, with the perpendicular legs 8 % low; against RANS the streamwise legs give the closest polars (drag within 0.3 % up to 10°, lift 0.879 against 0.869 at 10° where the ribs give 0.917).

The on-blade over Trefftz ratio is not a criterion between rib and streamwise legs (both are within

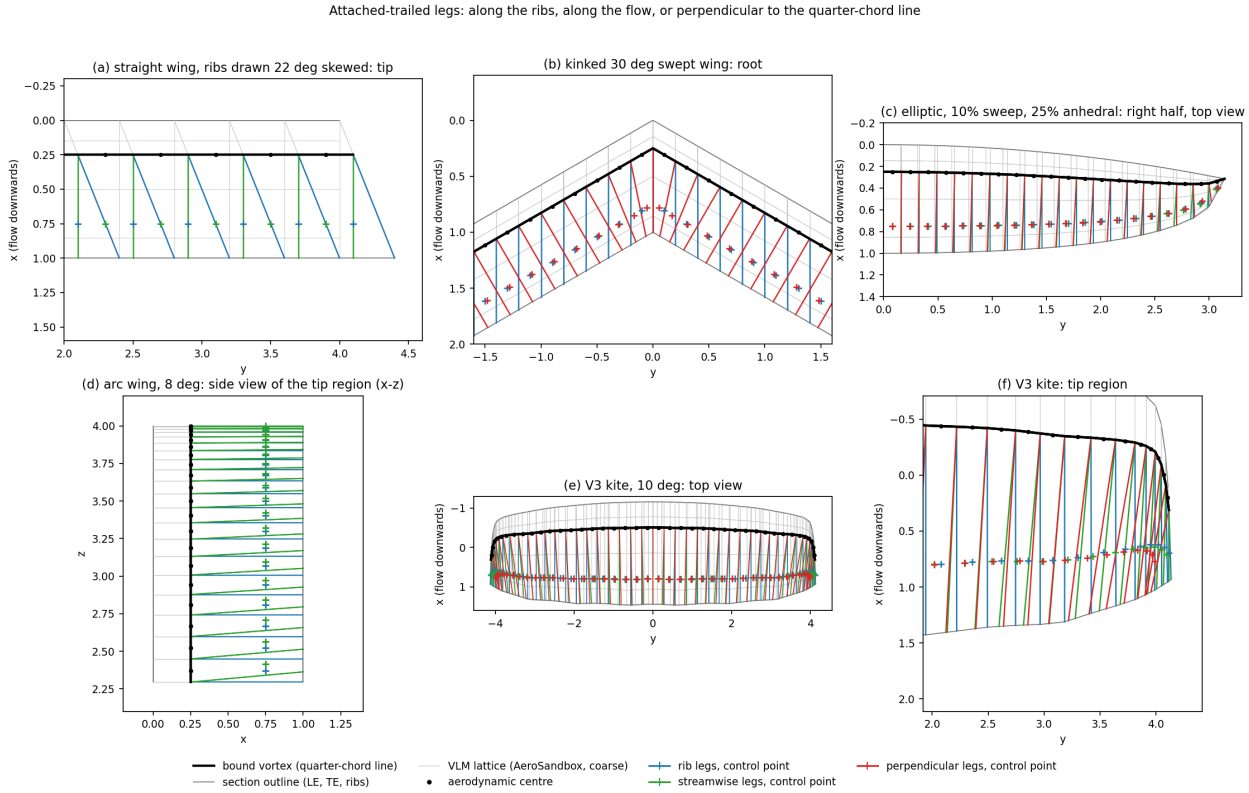


Figure 5: The three leg constructions. Top views (flow downwards) and one side view, with the bound vortex (black), the aerodynamic centres (dots), the control points (crosses in the colour of their legs) and, in grey, the lattice of the vortex-lattice referee. (a) A straight wing whose ribs are drawn 22° skewed: rib legs follow the drawing, streamwise and perpendicular legs end on the trailing-edge line and give the straight-wing result. (b) The root of the kinked 30° swept wing: streamwise ribs against the perpendicular fan. (c) The swept–anhedral elliptic wing. (d) The arc wing from the side: streamwise legs climb along the nearly vertical tip at the angle of attack. (e, f) The V3: the ribs are nearly streamwise over the middle and skewed at the tips, where the perpendicular legs become long (up to 2.9× the leg length at 80 panels) and end up to 1.6 c inboard of the section's own trailing-edge point.

Table 7: Top: on-blade over Trefftz induced drag with the attached-trailed force, $N = 80$ (`leg_direction_study.py`). Bottom: V3 viscous polars (CAD geometry, 2D RANS polars, 50 panels) against RANS [15], 1–13°; the drag error up to 10° in brackets.

	rib	streamwise	perpendicular
straight AR 8	1.000	1.000	1.000
straight, ribs skewed 22°	1.030	1.000	1.000
yawed 30°	1.002	1.002	0.991
kinked swept 30°	0.982	0.982	1.093
circular arc	0.999	1.035	0.999
elliptic, swept–anhedral	1.015	0.986	1.024
V3 kite, 10°	1.008	1.008	1.018 (crosses at $N = 160$)
V3 mean $\lvert\Delta C_L\rvert$	0.020	0.019	0.058
V3 mean $\lvert\Delta C_D\rvert/C_D$	3.0 % (1.4 %)	2.2 % (0.3 %)	3.5 % (2.5 %)

2 % except the streamwise arc at 1.035, where the climbing legs carry an attached-trailed force the flow-aligned legs of the other options do not); it does reject the perpendicular legs on the kinked wing (1.09, rising with N) and the singular earlier construction. The skewed-rib straight wing shows what the rib legs cost when the ribs are not aligned with the flow: the same wing, drawn with skewed ribs, gives 1.03 and 0.7 % less lift, whereas streamwise and perpendicular legs return the straight-wing result exactly.

Decision. The perpendicular legs are rejected with evidence. Between rib and streamwise legs the referee does not decide sharply: each is within 1 % of the VLM lift on four of the five wings it can judge, the ribs win on the arc and the streamwise legs on the elliptic wing and against RANS on the V3. The rib legs stay the default. The streamwise legs have the conceptual advantages: they depend only on the quarter-chord line, the trailing edge and the flow, not on how the input sections were drawn (the skewed-rib wing shows what that costs), they follow the inflow in sideslip, and on the real kite they give the closest polars to RANS. But the code's main use is trim finding with an inflow that varies along the span, where the streamwise construction can only use the mean direction and therefore is an approximation of its own idea, rebuilt at every inflow; the rib legs give a fixed vortex system of the same accuracy. The streamwise legs' known cost is on wings with nearly vertical tips, where the climbing legs lower the lift: on the arc wing −1.3, −2.1 and −3.0 % against the VLM at 4, 8 and 15° (rib legs: −0.2, +0.3, +1.6 %), while in induced drag at the same angles the streamwise legs are the closer ones (−1.3, −1.3, −1.0 % against −2.4, −3.5, −4.8 %); on the thesis curved-wing verification case the lift at high angle of attack drops by up to 7 %. On the V3 the streamwise lift slope (0.065 per degree) matches the RANS slope, the rib one (0.078) does not. An inflow without a mean direction (pure rotation) has no streamwise legs; the code then uses the rib legs for that inflow and logs a warning. The tables of Sections 2–4 and Figures 1–7 were made with rib legs; the quarter-chord regularisation itself is the same under either. Use `leg_direction="streamwise"` for uniform-inflow studies of wings whose sections are drawn obliquely to the flow, or in sideslip. The argument that chord and legs should follow the same direction has been tested in the one form the theory allows, the all-perpendicular CP1 construction, and it loses to the CP1 frame with flow-aligned legs; the paper itself separates the two choices.

The authors' numbers. For the swept–anhedral elliptic wing Table 2 of [1] gives $e_{\mathrm{Osw,Blade}} = 1.101$ against $e_{\mathrm{Osw,Trefftz}} = 1.080$ for LL-Gaunaa, i.e. on-blade induced drag 1.9 % *below* the Trefftz plane, and their VLM reference is 2.8 % below (1.106 against 1.075); the appendix [2] gives 5 % and 1 % below for the 15 % and 25 % swept wings. The authors read this as a limitation of the Trefftz-plane analysis for curved wings. Here the near-field drag of the VLM is likewise below its own Trefftz drag on the non-planar wings (8 % on the elliptic wing), for the same reason: trailing legs that are not aligned with the wind carry a force that the near-field evaluation drops.

4.5 Independent evidence: the arc self-induction

On the arc the chordwise self-induction that makes on-blade equal Trefftz is $u \approx -0.19\,\mathrm{m/s}$ (interpolating between the trailed-only value, ratio 1.079 at $u = 0$, and the on-line value at $N = 80$, ratio 0.951 at $u = -0.34\,\mathrm{m/s}$). The chordwise-spread sheet gives $u = -0.186\,\mathrm{m/s}$ at every N from 40 to 320 (Figure 1b). Nothing was tuned to obtain this agreement.

4.6 Effect on the V3 kite polars

Figure 7 and Table 8 show the V3 (CAD geometry, 2D RANS polars at Re 5×10^5, 50 panels, 2.82 m/s) with the old three-quarter-chord directions, the quarter-chord directions with the bound segments on the line and the regularisation, against RANS [15] and the wind-tunnel data of Poland et al. [15]. Lift is unchanged; the drag rises by a fraction of the induced drag, most at high lift and in sideslip, and lands on the RANS values up to 10°.

Table 8: V3 kite drag coefficient, $\beta = 0$.

α	old 3/4-chord	bound on the line	regularised	RANS (Re 10^6)	wind tunnel (Re 5×10^5)
5°	0.061	0.054	0.055	0.052 (4°)	0.076 (5.4°)
7°	0.075	0.063	0.065	0.064	0.090 (7.3°)
9°	0.094	0.074	0.079	0.084 (10°)	0.107 (9.4°)
13°	0.148	0.110	0.120	0.110	0.217 (13.4°)

The remaining gap to the wind tunnel is lift-dependent, $\Delta C_D \approx 0.003 + 0.037\,C_L^2$, and is shared by RANS ($C_D = 0.043 + 0.052\,C_L^2$ versus $0.041 + 0.085\,C_L^2$ measured). Poland et al. attribute it to the 3–4 mm model canopy, the flow behind the leading-edge tube and transition. It is not a lifting-line issue and the fix is not meant to close it; the old three-quarter-chord directions appeared to close part of it only because they overestimate the induced drag by 20–40 %.

5 Assessment and open items

- The mechanism of the fix (straighten locally, regularise the legs, keep the far field) is that of Reid and Hunsaker [7]; the difference-form goes back to Li et al. [6]. The contribution here is the replacement of the two empirical lengths by the thin-airfoil chordwise distribution, which fixes the cut-off at $0.15\,c$ from theory, and the demonstration that this yields Trefftz-consistent induced drag on curved kites. It should be presented as such, with [7, 5, 10] cited.
- To be confirmed with the authors of [1, 5]: whether their quarter-chord term is the curved-minus-straight bound-segment difference evaluated at the control point (Table 2), which reproduces their behaviour here with perpendicular legs, and how it behaves when the number of sections is doubled.
- With body rates the apparent velocity is built at the control points and the same array feeds the quarter-chord direction and the attached-trailed force; the rotational term's variation along the chord, $\omega \times (\mathbf{r}_{1/4} - \mathbf{r}_{3/4})$, is the effective torsional rate $T_0\dot{\theta}$ of Li, Gaunaa and Pirrung [4], whose coned-blade benchmark shows it worth 4 % of power at 10° cone. On the V3 at 20 m/s and a yaw rate of 0.3 rad/s the local angle error reaches 0.8° at the tips while the total drag changes 0.6 %, the error being antisymmetric across the span; the yawing moment is affected. To fix: evaluate the apparent velocity at the aerodynamic centre for the direction and at the leg points for the attached-trailed force.
- Settled with a referee (Section 4.4): the paper's perpendicular legs are rejected; rib legs stay the default, streamwise legs are available. A streamwise construction with the local inflow direction per node would remove the mean-direction limitation. Open: the attached-trailed force on legs that are not aligned with the flow (kinked wing 0.98, skewed ribs 1.03, streamwise

legs on the arc 1.035), and a V3 referee with the exact skewed tips (AeroSandbox unskews them). Not quantified: the section polars are rib-section polars while CP1 applies them in the span-perpendicular plane.

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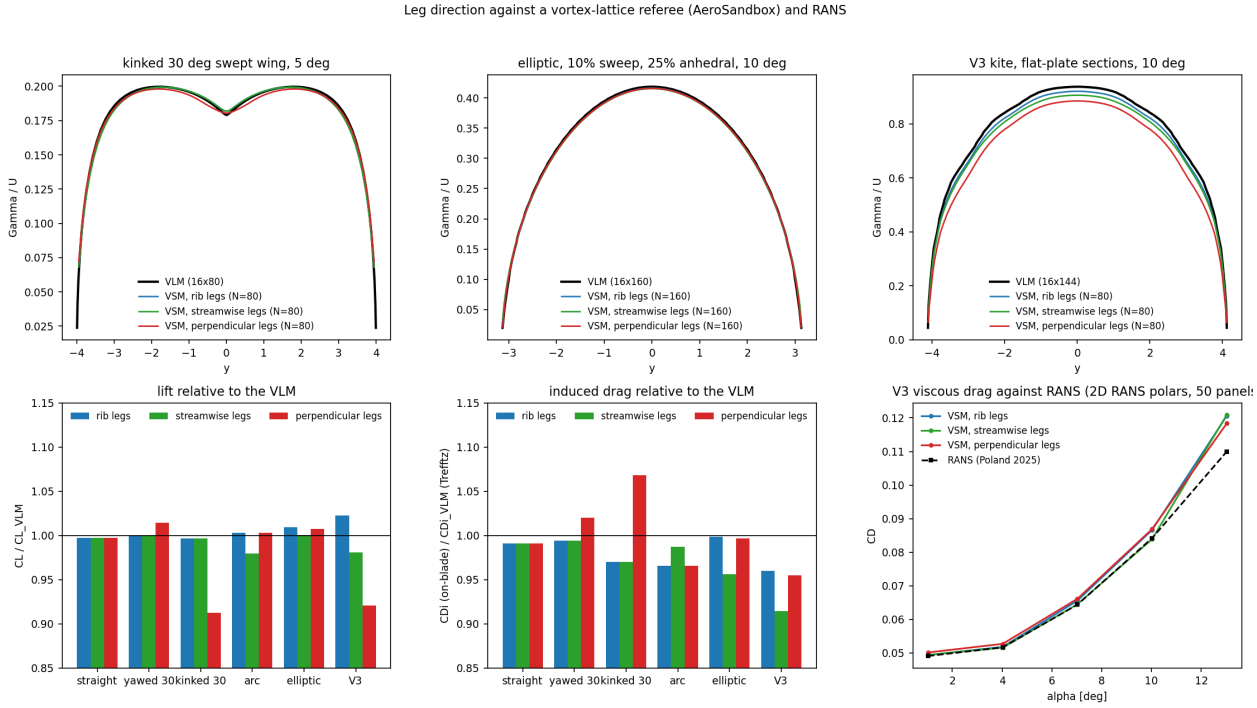


Figure 6: Leg direction against the vortex-lattice referee and RANS. Top: strip circulation of the VLM and of the VSM with the three leg options on the kinked swept wing, the elliptic wing and the inviscid V3. Bottom left and middle: lift and on-blade induced drag of the VSM relative to the VLM. Bottom right: V3 viscous drag against RANS [15] for the three options.

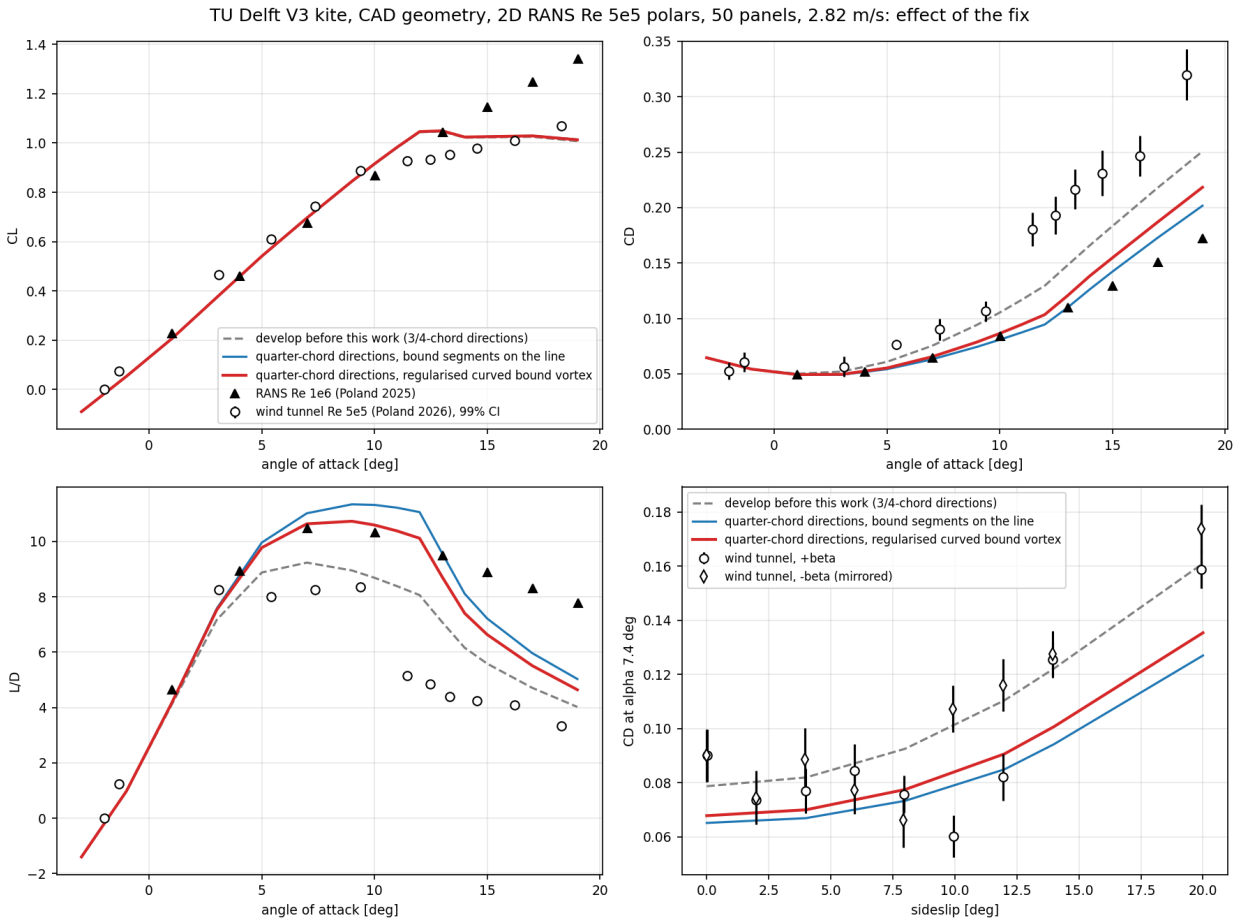


Figure 7: V3 kite polars (`docs/figures/09`) with the old three-quarter-chord directions, the quarter-chord directions with the bound segments on the line, and the regularisation, against RANS and the wind-tunnel data of Poland et al. (1:6.5 rigid model, no bridles, open-jet corrections applied).